

GPRI Global Plant LAI Database: A 30,000-species open dataset for quantifying urban greenery using the Green Plot Ratio metric

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Abstract

The Green Plot Ratio (GPR) is an ecological metric for quantifying three-dimensional urban greenery using Leaf Area Index (LAI), introduced by Ong (2003) and adopted in Singapore's Green Mark certification scheme. Despite its growing citation record (464+ citations as of 2026), no open, structured plant database has existed to support its consistent application across climate zones and urban typologies. This paper describes the GPRI Global Plant Database (version 1.0.0), a 34,429-species open dataset providing LAI values, botanical classification, biogeographic origin, and full data provenance for use in GPR calculation. The database integrates four source tiers — direct urban field measurement, open-ground scientific measurement, genus-level extrapolation, and Plant Functional Type inference — with each record carrying explicit confidence tier, measurement context, and AI-assistance flags. The dataset is freely accessible via a REST API (Supabase) and is the shared data layer for a suite of lightweight GPR calculation plugins for AutoCAD, Revit, Rhino, and Vectorworks. It is designed to improve as urban-context LAI measurement campaigns expand.

Background and Summary

Urban greenery delivers measurable ecosystem services — thermal regulation, stormwater attenuation, carbon sequestration, and biodiversity support — yet quantifying it in a form usable by architects and planners has remained methodologically inconsistent. Most green building rating systems assess greenery by area coverage alone, ignoring the three-dimensional structure of vegetation that determines actual ecological function.

The Green Plot Ratio (GPR), introduced by Ong (2003), addresses this by expressing greenery as a dimensionless ratio of total leaf area to site area:

$$\text{GPR} = \text{Total Leaf Area (m}^2\text{)} / \text{Site Area (m}^2\text{)}$$

where Total Leaf Area = sum of (LAI_i × Canopy Area_i) for all plant elements on site. Singapore's Green Mark scheme uses GPR thresholds as mandatory criteria for certified developments.

LAI — the ratio of one-sided leaf area per unit ground area — is the critical input. Existing databases (ORNL DAAC, TRY) were compiled for ecological and climate modelling, not urban design. Their measurements were taken in natural forests and plantations, and their data structures are not designed for per-species lookup by non-specialist users.

No open, structured database existed that: (1) provided LAI values queryable by species name; (2) covered the full range of plant categories used in urban landscape design; (3) disclosed the confidence and measurement context of each value; (4) was accessible via an API for integration into design tool plugins.

The GPRI Global Plant Database fills this gap. It is the shared data layer for a suite of lightweight plugins — GPR+AutoCAD, GPR+Revit, GPR+Rhino, and GPR+Vectorworks — each implementing the GPRSELECT workflow: the designer places their own plant objects in their CAD platform, the plugin reads canopy area and species name from those objects, queries the database for the LAI value, and calculates GPR. No geometry is generated by the plugin. The database is the product; the plugins are the lookup interface.

Methods

Overview of the construction pipeline

The database was constructed in four stages: (1) source acquisition, (2) source-specific processing into a normalised intermediate format, (3) priority-order merging with deduplication, and (4) schema validation and output. All processing scripts are written in Python and are included with the dataset.

The central design principle is tier priority: where multiple sources provide a value for the same species, the highest-confidence (lowest tier number) value is retained. No value is overwritten by a lower-confidence source.

Source 1 — Direct urban field measurement (Tier 1)

40 species. Source: Tan & Sia (2009).

The only currently available source of LAI values measured directly in an urban tropical context for GPR purposes. Tan and Sia (2009) measured LAI using a LAI-2000 Plant Canopy Analyser across 35 species in Singapore urban plantings, expressly for use in GPR calculation. 40 entries are recorded as Tier 1 in the current database, including five additional panel species confirmed against Singapore Green Mark technical documentation.

Context: Urban tropical Singapore. Climate: Koppen Af/Am. Method: LAI-2000 Plant Canopy Analyser (indirect optical).

Known limitation: 40 species covers the common tropical urban palette for Singapore. Perth, temperate, and arid-climate species are not represented in Tier 1.

Source 2 — Open-ground scientific measurement (Tier 2)

723 species. Sources: ORNL DAAC (Scurlock et al. 2001), TRY Plant Trait Database (Kattge et al. 2020).

Two peer-reviewed global LAI databases were downloaded and processed. ORNL DAAC provided 648 single-species entries covering global woody plants (1932-2011). TRY Plant Trait Database (v6.0) provided broader global coverage including some tropical genera absent from ORNL. The combined processing produced 723 validated Tier 2 entries.

Context: Natural forests, plantations, experimental plots — NOT urban. This limitation is recorded in the notes field of every Tier 2 entry.

Known limitation: Strongly biased toward temperate Northern Hemisphere species. Tropical urban species common in Asian and Australian cities are largely absent.

Source 3 — Genus-level mean extrapolation (Tier 3)

22 species. For species where no measured value exists but congeners are present in Tier 1 or Tier 2, a genus mean was calculated from available measured values. Intrageneric LAI variation is typically +/-20-30%. These entries are marked `ai_assisted = TRUE`.

Source 4 — Plant Functional Type (PFT) inference (Tier 4)

33,644 species. Source: GBIF Backbone Taxonomy + Bonan (2008) PFT LAI means.

The GBIF API was queried across 305 target genera. 33,432 accepted species names were returned. Each species was assigned a PFT code (Bonan 2008) based on growth form and native climate zone, with the published PFT LAI mean assigned as `lai_mean`.

PFT	Code	LAI mean	Range	Source
Tropical broadleaf evergreen tree	TrBE	5.1	4-8	Bonan 2008
Tropical broadleaf deciduous tree	TrBD	3.8	2-6	Bonan 2008
Temperate broadleaf deciduous tree	TeBD	4.0	2-6	Bonan 2008
Temperate needleleaf evergreen tree	TeNE	5.5	3-9	Bonan 2008
Shrub (all types)	MedS/TeDS	2.2	1-4	Bonan 2008
Grass / Groundcover	TrGr/TeGr	1.5	0.5-3	Bonan 2008
Palm	Palm	3.5	2-6	Running et al. 2000
Bamboo	Bamb	4.0	2-7	Literature estimate
Mangrove	MrBE	4.2	3-6	Literature estimate

All Tier 4 entries are marked `ai_assisted = TRUE`, `ai_model = claude-sonnet-4-6`.

Merging and deduplication

Sources are processed in tier priority order (Tier 1 to 4). Species matching is case-insensitive on the binomial name. Once a species is recorded from a higher-confidence source, all lower-confidence entries are discarded.

Output: `gpr_globalplantdb.csv` — 34,429 rows, 30 fields, 12.57 MB.

Schema design

The 30-field schema is organised into six sections:

- **Section 1 — Taxonomy:** Anchored to GBIF Backbone Taxonomy. gbif_taxon_key enables cross-referencing.
 - **Section 2 — Botanical classification:** Two parallel systems — growth_form (scientific controlled vocabulary) and landscape_category (design-practice vocabulary). This dual classification is the primary schema design contribution.
 - **Section 3 — Biogeography:** native_region and native_koppen support climate-zone filtering.
 - **Section 4 — LAI data:** lai_mean, lai_min, lai_max, lai_sd, lai_n, lai_method, lai_context, lai_measurement_koppen, pft.
 - **Section 5 — Provenance:** tier, tier_source, urban_context, sources, notes.
 - **Section 6 — Record metadata:** entry_date, data_version, ai_assisted, ai_model.
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Data Records

Property	Value
Format	CSV (UTF-8, comma-delimited)
Rows	34,429
Fields	30
File size	12.57 MB
Version	1.0.0
Entry date	2026-05-25

Distribution by tier:

Tier	Name	Count	Percent
1	Urban field measured	40	0.1%
2	Open-ground measured	723	2.1%
3	Genus mean	22	0.1%
4	PFT mean (inferred)	33,644	97.7%
Total		34,429	

Distribution by landscape category:

Category	Count
Tree	12,276
Shrub	9,530
Groundcover	4,802
Climber	2,698
Grass	2,201
Palm	1,403
Bamboo	900
REVIEW	403
Mangrove	210

Technical Validation

- **Binomial name validation:** All species names validated against standard binomial regex. 361 non-conforming entries assigned REVIEW category.
 - **LAI range plausibility:** Values not filtered by absolute range; lai_min/lai_max communicate uncertainty.
 - **Tier priority integrity:** No species appears more than once; no Tier 4 entry exists for a species present in Tier 1-3.
 - **AI-assistance flag coverage:** ai_assisted = TRUE: 33,426 entries; ai_assisted = FALSE: 1,003 entries; blank: 0.
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Usage Notes

Intended use

The GPRSELECT workflow is consistent across all four platforms (GPR+AutoCAD, GPR+Revit, GPR+Rhino, GPR+Vectorworks):

1. The designer places their own 3D plant objects in the CAD platform — no geometry is generated by the plugin.
2. The designer runs GPRSELECT and assigns a species name to each plant object.
3. The plugin queries the database for lai_mean and tier for each species.
4. The plugin calculates GPR and reports tier composition.

Urban LAI adjustment

All Tier 2 values were measured in natural or plantation conditions. Urban LAI is typically 30-60% lower. For regulatory applications, unadjusted values are appropriate as a conservative upper bound.

Limitations

1. **Urban vs natural LAI mismatch (Tier 2).** 723 species carry natural-context LAI values. Urban LAI may be 30-60% lower.
 2. **PFT inference uncertainty (Tier 4).** 33,644 species carry PFT mean values with wide uncertainty ranges.
 3. **Temperate bias.** Tier 2 data is dominated by temperate Northern Hemisphere species.
 4. **Canopy form classification.** Programmatic assignment is unreliable for non-standard genera.
 5. **Synonym resolution incomplete.** Exact binomial string matching only; synonym mapping incomplete in v1.0.0.
 6. **AI-assisted provenance.** Tier 3 and Tier 4 entries produced with AI assistance; disclosed via ai_assisted and ai_model fields.
 7. **Version 1.0.0 is a foundation.** Scientific value lies in schema design, provenance framework, and 763 Tier 1-3 entries.
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Future Development

- **v1.1:** Tier 1 expansion — Perth urban field LAI campaign (50-100 species).
 - **v1.2:** Tier 2 enrichment — AusTraits and NParks integration.
 - **v1.3:** Urban adjustment factors per installation type.
 - **v2.0:** Peer-reviewed methods paper on urban LAI adjustment framework.
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Code Availability

Script	Function
build_gpr_globalplantdb.py	Master build — merges all sources
download_gbif_comprehensive.py	GBIF API download (305 genera)
build_lai_global.py	Builds LAI_global.csv from Tier 1-3
lai_categorise.py	Assigns landscape categories
enrich_koppen.py	Populates native_koppen from GBIF
upload_to_supabase.py	Pushes to Supabase REST API
merge_singapore_lai.py	Processes Tan & Sia (2009) values

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Acknowledgements

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